SPY / SPX morning hedge

Three-year comparison · evidence through August 2025

Executive summary

January-August 2025 was the positive exception among the three historical samples. The same fixed morning hedge lost money after modeled costs in 2023 and 2024, including their January-August windows. A repeatable after-cost edge remains unproven.

• On the common January-August window, mean net P&L per package was -$56.12 in 2023 across 131 pairs, -$146.26 in 2024 across 131 pairs, and +$653.39 in 2025 across 127 pairs.

• Full-year 2024 averaged -$90.03 after modeled costs, with a -$57.16 median and 39.0% positive packages. The hedge improved the mean by $418.07 versus no hedge, which averaged -$508.10.

• Average overnight stock contribution was -$1.90 in full-year 2024 versus +$777.20 in January-August 2025. Favorable overnight moves explain the positive 2025 result; its average modeled costs were higher.

• Removing the ten largest 2025 winners leaves 117 observations with a -$105.30 mean. The original 2025 median was -$10.08. This is a post-hoc sensitivity, not a trading filter.

Cohort results

Focal case: full-cross fills, base costs, gross prefunding, 1 bp stock slippage. The original formal primary arm was fixed-500; this morning arm was secondary.

Cohort	n	Mean	Median	Total	Positive	Costs	Overnight
Full 2023	196	-$48.96	-$66.70	-$9,596.60	40.3%	$21,049.03	$16,260.00
Full 2024	195	-$90.03	-$57.16	-$17,555.09	39.0%	$26,792.03	-$370.00
Jan-Aug 2025	127	$653.39	-$10.08	$82,980.62	48.0%	$19,828.50	$98,705.00
Jan-Aug 2023	131	-$56.12	-$47.16	-$7,351.92	41.2%	$13,703.35	$9,310.00
Jan-Aug 2024	131	-$146.26	-$65.13	-$19,160.49	38.2%	$17,970.43	-$8,380.00

Mean, median, and total in the table have distinct units: mean and median are USD per expiry; total and component figures are cohort USD totals. The Markdown companion gives all six costs and the diagnostic Net minus overnight, which retains modeled costs and is a total USD value, not a tradable return.

Components, states and tails

Cohort	Option base	q0 mean	Dividends	q0 / q1000	Assigned	Residual L/S/0
Full 2023	-$1,663.00	-$124.54	-$3,144.57	95 / 101	104	33 / 36 / 127
Full 2024	$8,012.00	-$508.10	$1,594.94	77 / 118	114	27 / 23 / 145
Jan-Aug 2025	$2,343.00	$191.97	$1,761.12	52 / 75	69	27 / 21 / 79
Jan-Aug 2023	$186.00	-$58.50	-$3,144.57	66 / 65	68	23 / 26 / 82
Jan-Aug 2024	$5,595.00	-$505.49	$1,594.94	50 / 81	78	20 / 17 / 94

Top and bottom tail removals

Cohort	Top 1	Top 5	Top 10	Bottom 1	Bottom 5	Bottom 10	n left
Full 2023	-$74.66	-$143.47	-$218.13	-$27.90	$43.58	$122.34	186
Full 2024	-$138.85	-$245.66	-$327.82	-$52.86	$60.80	$157.81	185
Jan-Aug 2025	$497.67	$152.59	-$105.30	$736.85	$981.57	$1,117.78	117
Jan-Aug 2023	-$94.73	-$195.90	-$306.04	-$24.58	$79.13	$181.65	121
Jan-Aug 2024	-$200.08	-$324.69	-$401.54	-$91.24	$78.94	$180.56	121

Tail means are USD per package after removing the largest or smallest k net outcomes. n left is the count after removing 10. This post-hoc sensitivity check is not a trading filter.

Monthly net results

Month	2023 n / mean	2024 n / mean	2025 n / mean
01	15 / -$317.49	16 / -$0.20	14 / -$107.17
02	15 / $207.01	16 / -$470.75	15 / $816.54
03	19 / -$291.41	16 / -$462.13	16 / $859.12
04	15 / $660.30	17 / -$356.23	17 / $1,017.19
05	17 / -$165.66	18 / -$8.87	17 / $1,612.49
06	17 / -$424.32	14 / -$550.27	14 / $784.83
07	14 / -$131.94	16 / $626.51	18 / $171.21
08	19 / $95.49	18 / -$18.66	16 / -$17.94
09	16 / -$856.56	15 / $602.48	0 / n/a
10	17 / $542.12	19 / -$251.43	0 / n/a
11	16 / $11.85	15 / -$572.48	0 / n/a
12	16 / $128.42	15 / $395.51	0 / n/a

Costs and uncertainty

Source-cohort 95% type-7 intervals

Source cohort	Hedge mean, absolute interval	Paired difference vs q0
2023 full year	-$48.96 [-$221.01, $117.67], n=196	$75.58 [-$119.83, $273.89], n=196
2024 full year	-$90.03 [-$279.89, $99.65], n=195	$418.07 [$132.92, $714.32], n=195
2025 Jan-Aug	$653.39 [$125.64, $1,274.85], n=127	$461.42 [-$90.96, $1,063.93], n=127

Intervals are source-provided for 2023 full year, 2024 full year, and 2025 Jan-Aug. These are previously inspected historical samples, not holdouts or predictive validation. Filtered Jan-Aug 2023 and Jan-Aug 2024 cohorts have no computed intervals. No common-window confidence intervals or between-year p-values were computed.

Six modeled cost components

Cohort	Option fees	Stock fees	Borrow	Premium funding	Hedge funding	Slippage
Full 2023	$1,401.40	$850.00	$2,102.06	$1.96	$9,414.52	$7,279.09
Full 2024	$1,394.25	$840.00	$1,628.99	$3.34	$13,836.20	$9,089.25
Jan-Aug 2025	$908.05	$615.00	$1,331.59	$1.65	$9,619.94	$7,352.28
Jan-Aug 2023	$936.65	$570.00	$1,438.33	$1.28	$5,957.15	$4,799.94
Jan-Aug 2024	$936.65	$590.00	$1,146.63	$1.93	$9,121.29	$6,173.93

36-case sensitivity per year

Each sensitivity mean is net USD per expiry; the six cost component figures above are cohort totals in USD.

Year	Fill	Cost	Slip	Funding	n	Mean
2024	full_cross	base	0.0 bp	assignment cash net	195	$13.54
2024	full_cross	base	0.0 bp	gross prefunding	195	-$43.41
2024	full_cross	base	1.0 bp	assignment cash net	195	-$33.08
2024	full_cross	base	1.0 bp	gross prefunding	195	-$90.03
2024	full_cross	base	5.0 bp	assignment cash net	195	-$219.52
2024	full_cross	base	5.0 bp	gross prefunding	195	-$276.47
2024	full_cross	stress	0.0 bp	assignment cash net	195	-$33.64
2024	full_cross	stress	0.0 bp	gross prefunding	195	-$147.54
2024	full_cross	stress	1.0 bp	assignment cash net	195	-$80.25
2024	full_cross	stress	1.0 bp	gross prefunding	195	-$194.15
2024	full_cross	stress	5.0 bp	assignment cash net	195	-$266.69
2024	full_cross	stress	5.0 bp	gross prefunding	195	-$380.59
2024	halfway	base	0.0 bp	assignment cash net	195	$23.84
2024	halfway	base	0.0 bp	gross prefunding	195	-$33.11
2024	halfway	base	1.0 bp	assignment cash net	195	-$22.77
2024	halfway	base	1.0 bp	gross prefunding	195	-$79.72
2024	halfway	base	5.0 bp	assignment cash net	195	-$209.21
2024	halfway	base	5.0 bp	gross prefunding	195	-$266.16
2024	halfway	stress	0.0 bp	assignment cash net	195	-$23.33
2024	halfway	stress	0.0 bp	gross prefunding	195	-$137.23
2024	halfway	stress	1.0 bp	assignment cash net	195	-$69.94
2024	halfway	stress	1.0 bp	gross prefunding	195	-$183.84
2024	halfway	stress	5.0 bp	assignment cash net	195	-$256.38
2024	halfway	stress	5.0 bp	gross prefunding	195	-$370.28
2024	midpoint	base	0.0 bp	assignment cash net	195	$34.15
2024	midpoint	base	0.0 bp	gross prefunding	195	-$22.80

Year	Fill	Cost	Slip	Funding	n	Mean
2024	midpoint	base	1.0 bp	assignment cash net	195	-$12.46
2024	midpoint	base	1.0 bp	gross prefunding	195	-$69.41
2024	midpoint	base	5.0 bp	assignment cash net	195	-$198.90
2024	midpoint	base	5.0 bp	gross prefunding	195	-$255.85
2024	midpoint	stress	0.0 bp	assignment cash net	195	-$13.02
2024	midpoint	stress	0.0 bp	gross prefunding	195	-$126.92
2024	midpoint	stress	1.0 bp	assignment cash net	195	-$59.63
2024	midpoint	stress	1.0 bp	gross prefunding	195	-$173.53
2024	midpoint	stress	5.0 bp	assignment cash net	195	-$246.07
2024	midpoint	stress	5.0 bp	gross prefunding	195	-$359.97
2025	full_cross	base	0.0 bp	assignment cash net	127	$760.59
2025	full_cross	base	0.0 bp	gross prefunding	127	$711.28
2025	full_cross	base	1.0 bp	assignment cash net	127	$702.70
2025	full_cross	base	1.0 bp	gross prefunding	127	$653.39
2025	full_cross	base	5.0 bp	assignment cash net	127	$471.13
2025	full_cross	base	5.0 bp	gross prefunding	127	$421.82
2025	full_cross	stress	0.0 bp	assignment cash net	127	$695.48
2025	full_cross	stress	0.0 bp	gross prefunding	127	$596.86
2025	full_cross	stress	1.0 bp	assignment cash net	127	$637.58
2025	full_cross	stress	1.0 bp	gross prefunding	127	$538.97
2025	full_cross	stress	5.0 bp	assignment cash net	127	$406.02
2025	full_cross	stress	5.0 bp	gross prefunding	127	$307.41
2025	halfway	base	0.0 bp	assignment cash net	127	$772.89
2025	halfway	base	0.0 bp	gross prefunding	127	$723.59
2025	halfway	base	1.0 bp	assignment cash net	127	$715.00
2025	halfway	base	1.0 bp	gross prefunding	127	$665.69
2025	halfway	base	5.0 bp	assignment cash net	127	$483.43
2025	halfway	base	5.0 bp	gross prefunding	127	$434.13
2025	halfway	stress	0.0 bp	assignment cash net	127	$707.78
2025	halfway	stress	0.0 bp	gross prefunding	127	$609.17
2025	halfway	stress	1.0 bp	assignment cash net	127	$649.89
2025	halfway	stress	1.0 bp	gross prefunding	127	$551.28
2025	halfway	stress	5.0 bp	assignment cash net	127	$418.32
2025	halfway	stress	5.0 bp	gross prefunding	127	$319.71
2025	midpoint	base	0.0 bp	assignment cash net	127	$785.20
2025	midpoint	base	0.0 bp	gross prefunding	127	$735.89
2025	midpoint	base	1.0 bp	assignment cash net	127	$727.30
2025	midpoint	base	1.0 bp	gross prefunding	127	$678.00
2025	midpoint	base	5.0 bp	assignment cash net	127	$495.74
2025	midpoint	base	5.0 bp	gross prefunding	127	$446.43
2025	midpoint	stress	0.0 bp	assignment cash net	127	$720.09
2025	midpoint	stress	0.0 bp	gross prefunding	127	$621.48
2025	midpoint	stress	1.0 bp	assignment cash net	127	$662.19
2025	midpoint	stress	1.0 bp	gross prefunding	127	$563.58
2025	midpoint	stress	5.0 bp	assignment cash net	127	$430.63
2025	midpoint	stress	5.0 bp	gross prefunding	127	$332.02

Method and provenance

Expiry morning SPY open selects quantity using the frozen selected strike + $0.01 rule, with 1e-10 tolerance. The modeled stock purchase occurs at expiry close; the residual closes at the next session open. Options use one-calendar-day DTE from entry close to expiry close. The stock holding can span weekends. The option package is 10 short SPY and one long SPXW, multiplier 100.

The original fixed-500 arm was the formal primary arm; this morning arm was secondary. These are previously inspected historical samples, not holdouts or predictive validation. The 2024 cohort covers expiries 2024-01-03 through 2024-12-31, with final next-session cover open on 2025-01-02. No rule tuning or outcome-based date selection is used.

15:59 quote-clock verification is unavailable for 2023 and verified for 2024/2025. Later closing references have execution limits. Trades are daily theoretical fills and do not establish operational edge. These are previously inspected historical samples, not holdouts or predictive validation.

Provenance hashes

Aggregate SHA-256 aa4887efd7e11f2478cb361b5ec0af37c863c324581cee2c4e3f17ff1f58aff6 · run receipt SHA-256 f2bc5ec4f7882e0c4339384d82ed4943852a5ee4d8be2e481a3167df032b8004

Comparison builder Git SHA 1ce2763318f930f793da29293f6b5df482ef3b42 · builder file SHA-256 c154660c93f5faa269742428d516cf0cd544232147eaa7a7af390a4363ce1047

Source evaluator Git SHAs: 2023 64e72887b782fdf838257a09877ffb817dfbfbdf · 2024/2025 44ed294deddb85df594ad1e1c8a5a91bb22d08d7

Checker SHA-256 b2ec4d5000533c1c492a7453a1c9c7563c619cf3c7dff73099930fc0ceb7ad83 · input manifest SHA-256 dac4f37eb6f1685308067cbffbca0b91a46ea79d286935e6f1874b61c5f36aa5 · protocol SHA-256 207577186b5c0c9c078dcc3f203fefa86558a793fe4cb3d5aa96838187c76642